

Khoa Minh Pham

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Education

Bucknell University

B.S. in Computer Science, Data Science Co-Major / GPA: 3.34

Lewisburg, PA

May 2027

Experience

Interactive Developer Intern

JPL Integrated Communications

Harrisburg, PA

Jun 2026 – Present

- Shipped and debugged responsive web features across 4+ client projects, resolving 5+ WordPress/PHP issues/features through implementation, stakeholder review, and QA.
- Contributed to an Electron desktop application backed by Node.js, Express, WebSockets, and Docker.

Software Engineering Consultant — Micro-Internship

Parker Dewey

Lewisburg, PA

Feb 2026 – Apr 2026

- Shipped a payment-gated cybersecurity risk assessment platform using Next.js, FastAPI, PostgreSQL, Clerk, Stripe, and the Groq API, supporting authenticated assessments and monetized AI-generated reports.
- Built a weighted scoring engine across eight security domains, converting readiness signals into 0–100 scores and Low-to-Critical risk tiers.

Student Software Developer — Academic Chatbot

Bucknell University

Lewisburg, PA

Oct 2025 – May 2026

- Built an authenticated academic RAG chatbot using Next.js, FastAPI, ChromaDB, and OpenAI APIs, enabling users to query documents and receive grounded, source-linked answers.
- Engineered a document ingestion pipeline for PDFs, Markdown, and Google Docs with metadata-based indexing for traceable retrieval.
- Improved multi-turn response quality through query classification, conversation summarization, and user-context extraction.

Student Software Developer — KALMUS Web

Bucknell University

Lewisburg, PA

Oct 2025 – May 2026

- Built a Next.js, TypeScript, and Python application for computational film-color analysis, connecting browser workflows to SLURM/HPC jobs through NFS storage and asynchronous job-state tracking.
- Implemented chunked video uploads with four concurrent requests and 5–25 MB chunks, reducing large-file upload time by 30–50%.

Projects

SpotOn — AI Music Research Platform

React, TypeScript, Groq, Spotify, Vercel, Redis

- Built and deployed an AI-powered music research platform with Spotify OAuth, real-time catalog search, direct-to-account playlist saving, and sourced reports covering emotional and sonic fingerprint, credits, and cultural context.
- Engineered a tool-calling agent with up to 6 research rounds and 2 live web searches per report, streaming progress through SSE and caching completed results in Upstash Redis.

FeelBit — Full-Stack Mood Tracking Platform

Java, Spring Boot, React, PostgreSQL, Docker, Gemini

- Rebuilt a JavaFX project as a containerized Spring Boot and React platform with PostgreSQL, Flyway, mood analytics, and cached AI-powered wellness suggestions.
- Implemented 1-hour JWT access tokens and rotating 7-day refresh tokens stored as SHA-256 hashes, with BCrypt authentication and automatic session restoration.

Secure Client-Server Authentication and RBAC

C, OpenSSL, Makefile

- Implemented client-server authentication with SHA-256 hashing, RSA key generation, replay-attack protection, and AES-256-CBC sessions secured with HKDF-derived keys.

Skills

Languages: Python, TypeScript, JavaScript, Java, C, SQL, PHP

Frameworks: React, Next.js, FastAPI, Node.js, Express, Spring Boot

AI and Data: RAG, LLM APIs, ChromaDB, PostgreSQL, SQLite

Infrastructure and Security: Docker, AWS, GitHub Actions, OpenSSL, SLURM, WebSockets, RBAC